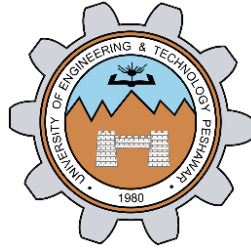


Project Report

Smart Security System



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CSE-303L Data Communication & Networks Lab

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Class Section: C

“On ours honor, as students at University of Engineering and Technology, We
have neither given nor received unauthorized assistance on this academic
work.”

Submitted to:

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June 24, 2024

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1. Abstract:

With the increasing global incidence of theft, the demand for effective security solutions has increased. This project focuses on developing a smart security system designed to protect homes, offices, rooms, or any restricted areas from unauthorized entry. Implemented using Packet Tracer, a robust network simulation tool, the system leverages IoT devices to significantly enhance security measures. Additionally, the project includes the implementation of remote connections and a DNS server, further extending the system's capabilities.

Keywords: Anti-Theft system, Smart Security System, Packet Tracer, IoT application, Remote Connection

2. Problem Statement:

The objective of this project is to develop a smart security system capable of detecting intrusions using IoT devices. The system will incorporate DNS access and remote connections to enhance its functionality. Implementing this project in Packet Tracer provides a simulated environment to test and refine the system before real-world deployment.

3. Introduction:

This project focuses on developing a smart security system designed to protect various premises, such as homes, offices, and restricted areas, from unauthorized entry. Utilizing Packet Tracer, a robust network simulation tool, we have created a comprehensive security system that integrates multiple IoT components to monitor and secure premises effectively.

Our system employs devices such as IoT doors, motion detectors, webcams, siren alarms, and home gateways, all interconnected to provide real-time monitoring and response to potential security threats. By incorporating remote connections and a DNS server, we have extended the system's capabilities, allowing users to access and control the security features from remote locations. This project not only demonstrates the practical application of IoT in enhancing security measures but also provides a simulated environment to test and refine the system before real-world deployment. Through this innovative approach, we aim to showcase the significant potential of IoT in developing advanced, user-friendly security solutions that address the growing need for effective anti-theft systems.

4. Methodology:

The methodology involved setting up Packet Tracer, adding and configuring network components, creating test scenarios, testing the system's functionality, and extending its capabilities for remote access. This structured approach ensured the systematic development and testing of the smart security system. Steps in our methodology are as follows:

- Setting Up Packet Tracer.
- Adding Components and configuring Devices.

- Creating Scenarios and programming Logic.
- Working of Smart Security System.
- Extending the system's capabilities.

a. Setting Up Packet Tracer:

Setting up Packet Tracer involved downloading the software from the Cisco Networking Academy website. After downloading, we initiated the installation process by running the installer file. Following the on-screen instructions, we selected the installation directory and agreed to the terms and conditions. Once installed, we launched Packet Tracer to begin creating and simulating network topologies.

b. Adding Components and configuring Devices:

We first add the server, pc and a switch gave it appropriate name and assigned IP addresses.

pc → 10.10.10.6/8

server → 10.10.10.1/8

We connected these two with the switch and made our first LAN named as LAN1 as shown in figure 1. Note that 10.10.10.10/8 is the IP address of router which we will connect later.

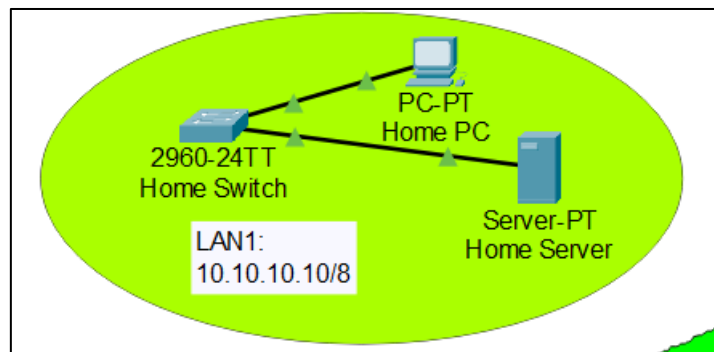


Figure 1, LAN1

In the home server we made the IoT Service on and registered new account by going to web browser and entering server IP which is 10.10.10.1 and signed up with username admin and password 1234 as shown in figure 2.

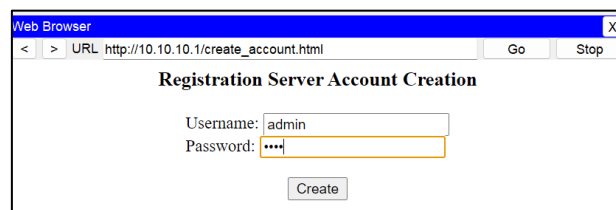


Figure 2, configuring home server

Then we added door, siren also known as alarm, motion detector, webcam, smartphone and home gateway to our workspace. We configured the home gateway LAN and Wireless port with IP addresses as follows:

Home Gateway LAN → 192.168.25.1

Wireless → Authentication: WPA2-PSK and Password: 12345678

We connected our other IoT devices one by one using wirelessly into our home server by selecting remote server in IoT section in config tab and entering values for server addresses, username and password for the door named as Entry Door the values are shown in figure 3.

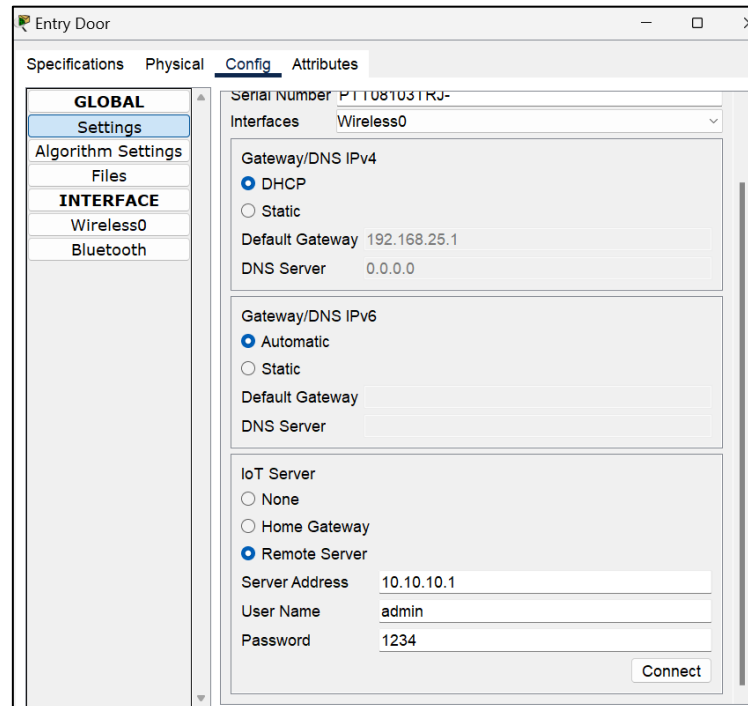


Figure 3, Entry Door IoT Device

these IoT devices will be connected to home gateway wirelessly as shown in figure 4. And thus, we made LAN2 as shown in figure 4.

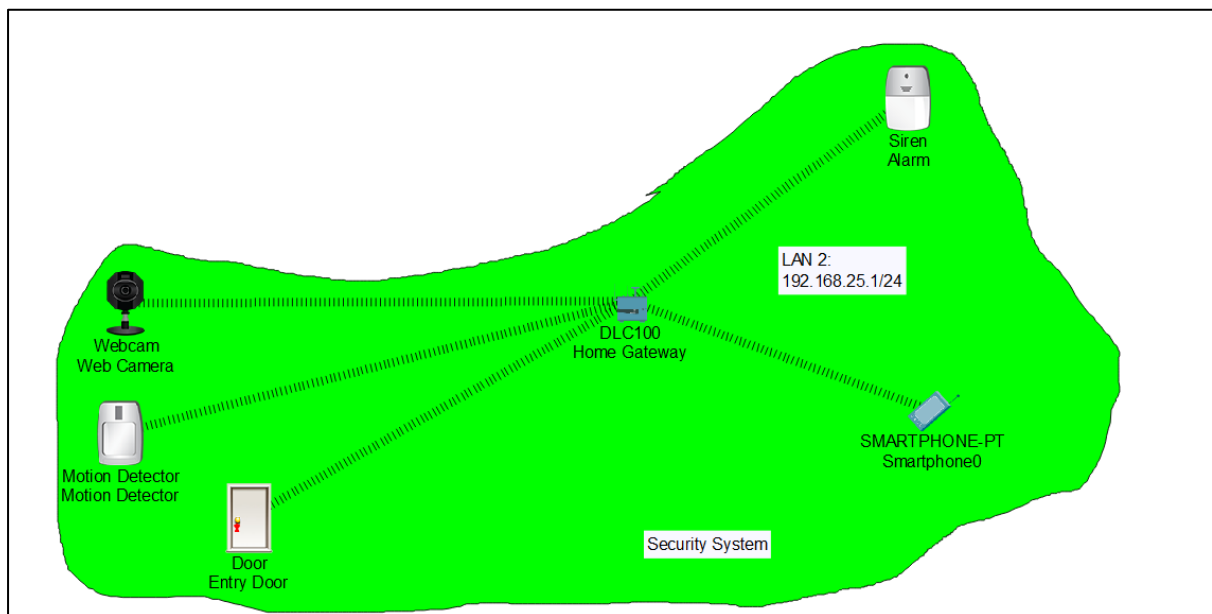


Figure 4, LAN2

We connected the LAN1 and LAN2 by connecting the link between home switch and home gateway internet port. Assigning internet port IP 10.10.10.100/8. We also added router named as home router to switch ethernet port and assigning home router FastEthernet0/0 port IP 10.10.10.10/8. This home router will become useful when we will establish remote connection to our security system. Figure 5 shows the connection between LAN1, LAN2 and home router

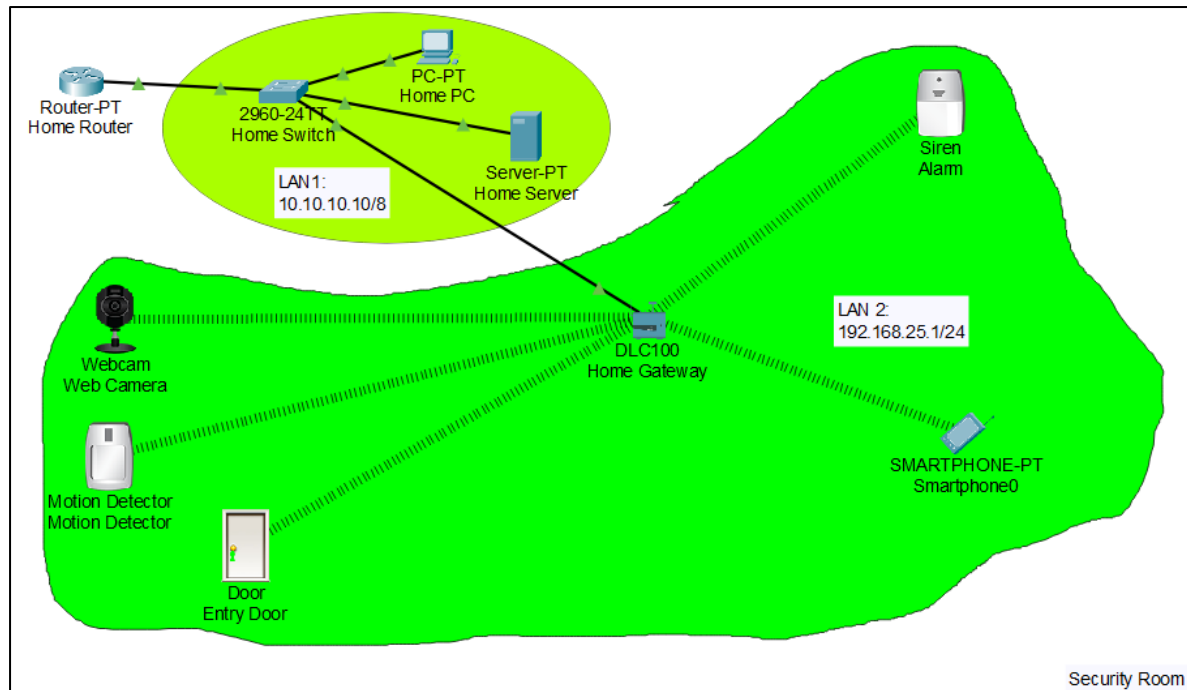


Figure 5, Connection between LAN1, LAN2 and home router

We accessed the home server by smartphone using web browser and IP 10.10.10.1 and the result is shown in figure 5.1, showing all the IoT devices connected to our home server.

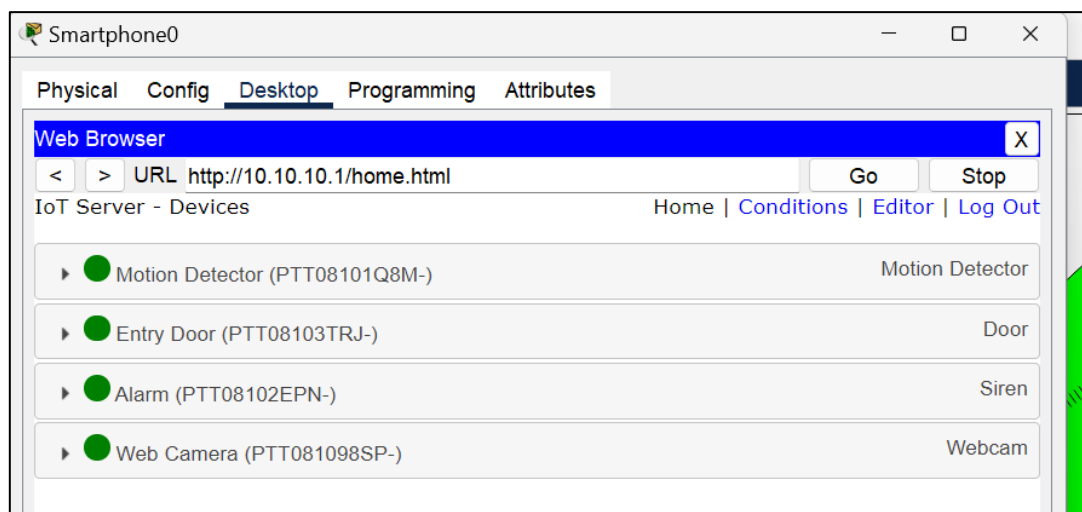


Figure 5.1, accessing home server by web browser on smartphone

c. Creating Scenarios and programming Logic:

We can manually activate and deactivate these IoT using access to the home server. And we can also automate these using conditions. We have created two scenarios as follows:

i. No one detected:

When no person is around the door, the motion detector is off, the webcam is off, and there is no alarm as shown in figure 6.

ii. Intruder detected:

When a person is detected moving at the door through the motion detector, the webcam activates to monitor the door, and the alarm sounds. We can lock and unlock the door for entry to our security room as shown in figure 6.

Actions	Enabled	Name	Condition	Actions
Edit Remove	Yes	Intruder Detected	Motion Detector On is true	Set Web Camera On to true Set Alarm On to true
Edit Remove	Yes	No one Detected	Motion Detector On is false	Set Web Camera On to false Set Alarm On to false

[Add](#)

Figure 6, Scenarios and Device Conditions

d. Working of Smart Security System:

We access the home server by home PC and test the two scenarios. We can also access the home server from smartphone.

i. Scenario 1: No one detected:

Now as we can see from figure 7, that motion detector is off as well as webcam and alarm. The webcam shows black screen indicating webcam is off.

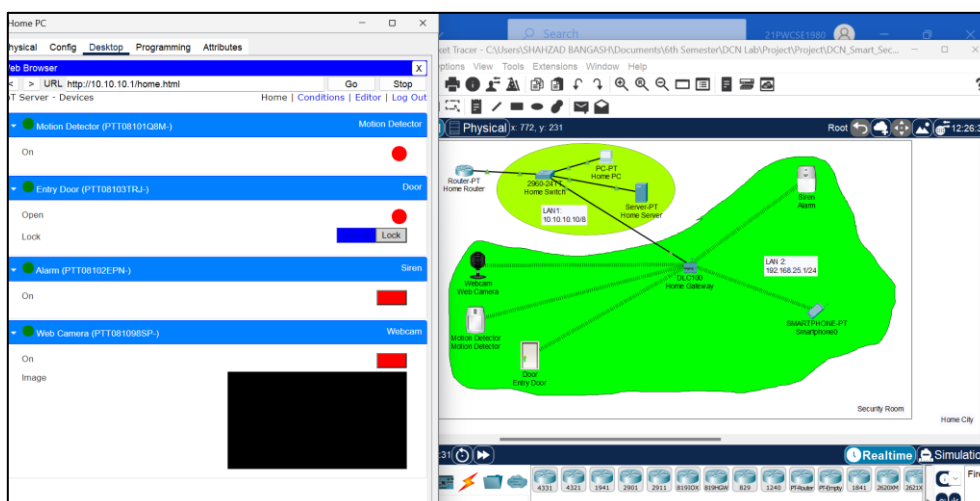


Figure 7, No one detected scenario

ii. Scenario 2: Intruder detected:

Now by pressing the ALT button and moving the mouse through motion detector we can trigger the motion detector and make it on, and we can see from figure 8, that the webcam is on, showing the person which is detected by motion detector and the alarm is also on and therefore we can also lock and unlock the door manually to let the person enter or not.

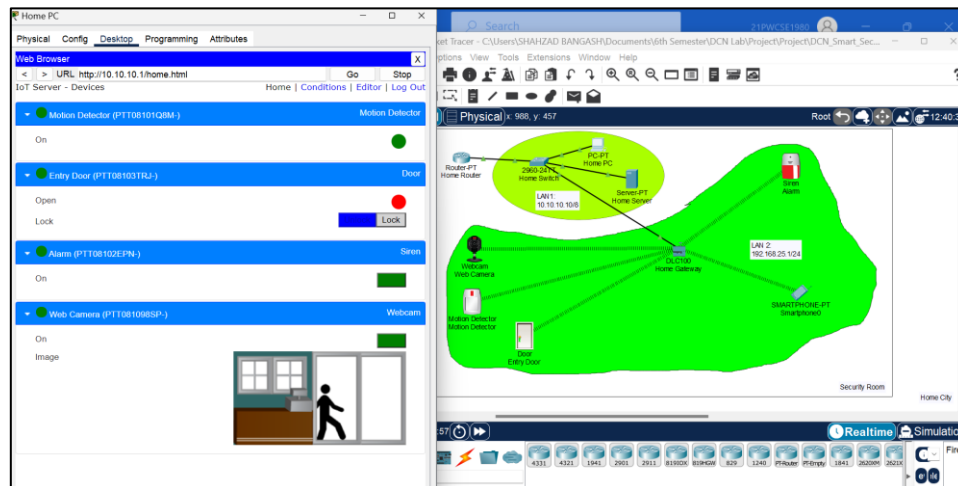


Figure 8, Intruder detected scenario

e. Extending the system's capabilities:

We extended the smart security system more by accessing it remotely from another building and even from another city. We also implemented the DNS server so we can access home server by IP address and as well as from URL which is www.home.com.

i. Implementing Remote Connection from Another Building:

Figure 9 and 10 shows that we created a remote office building for remote connection to our smart security system. This remote office has one LAN named LAN3 and an office router. LAN3 includes an Office Switch, Office PC, and Office Laptop connected to the switch, which is connected to the office router. The IP address of LAN3 is 192.168.10.0/24. The office router connects to the home router using a serial cable on port Serial 2/0, with an IP address of 172.168.30.30/16. This setup allows for remote connections to the home router.

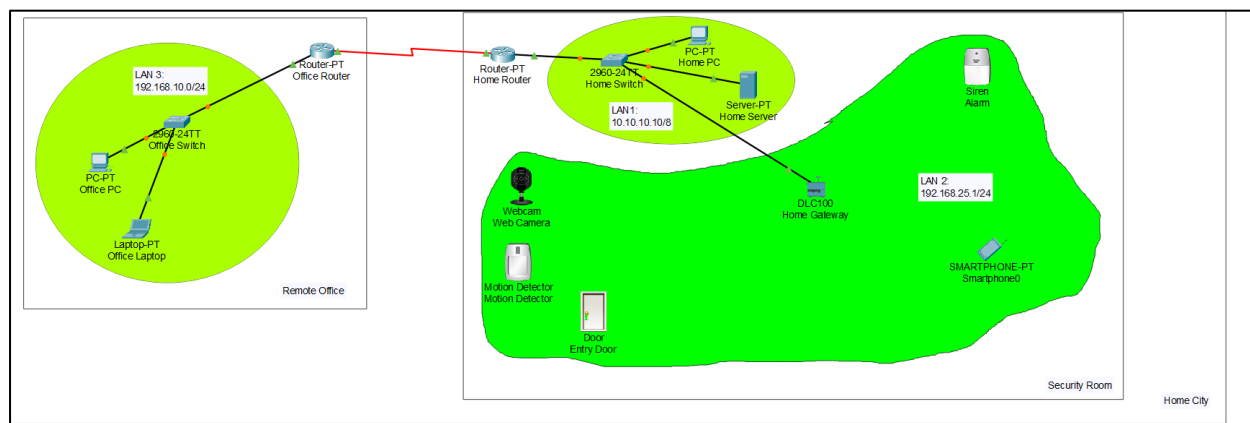


Figure 9, Remote Connection from Remote Office Building

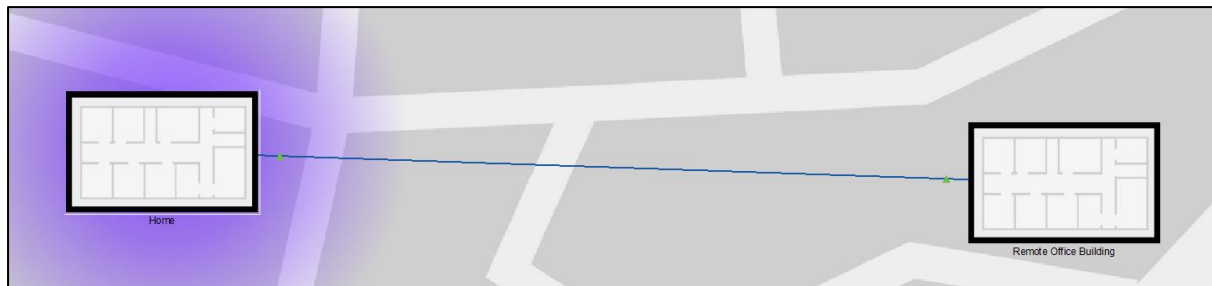


Figure 10, Remote Connection from Remote Office Building in physical topology

ii. Implementing Remote Connection from Another City:

Figure 11, 12 and 13 shows that we extended our connection to another city named Far City, which has one building named Far Office with two departments. The home city and far city are connected using home and far city routers via fiber optical cable.

Department 1:

LAN4 includes one switch (DP1 Switch), two PCs (DP1_PC and DP2_PC), and one printer (DP1_Printer). The hosts connect to the DP1 Switch, which connects to the DP1 Router. The DP1 Router connects serially to the DP2 Router and Far City Router.

Department 2:

LAN5 and LAN6 have similar setups. LAN5 includes one switch, two PCs, one printer, and one router. LAN6 includes one switch and one DNS server, connected to the DP2 Router, which connects to DP1 Router and Far City Router for cross-city connections. The IP addresses are sub netted from into two subnets, each supporting up to 254 hosts.

Calculation for subnetting IP 192.168.254.0/23:

Subnet 1:

Network address: 192.168.254.0

Subnet mask: 255.255.255.0 (/24)

Usable IP range: 192.168.254.1 to 192.168.254.254

Broadcast address: 192.168.254.255

Subnet 2:

Network address: 192.168.255.0

Subnet mask: 255.255.255.0 (/24)

Usable IP range: 192.168.255.1 to 192.168.255.254

Broadcast address: 192.168.255.255

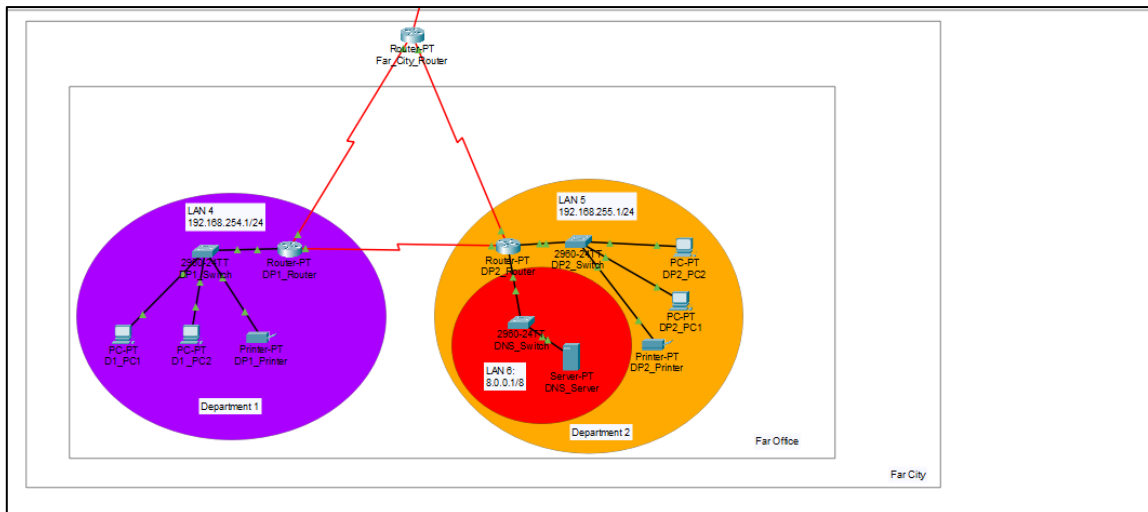


Figure 11, Far Office Structure in Far City Building

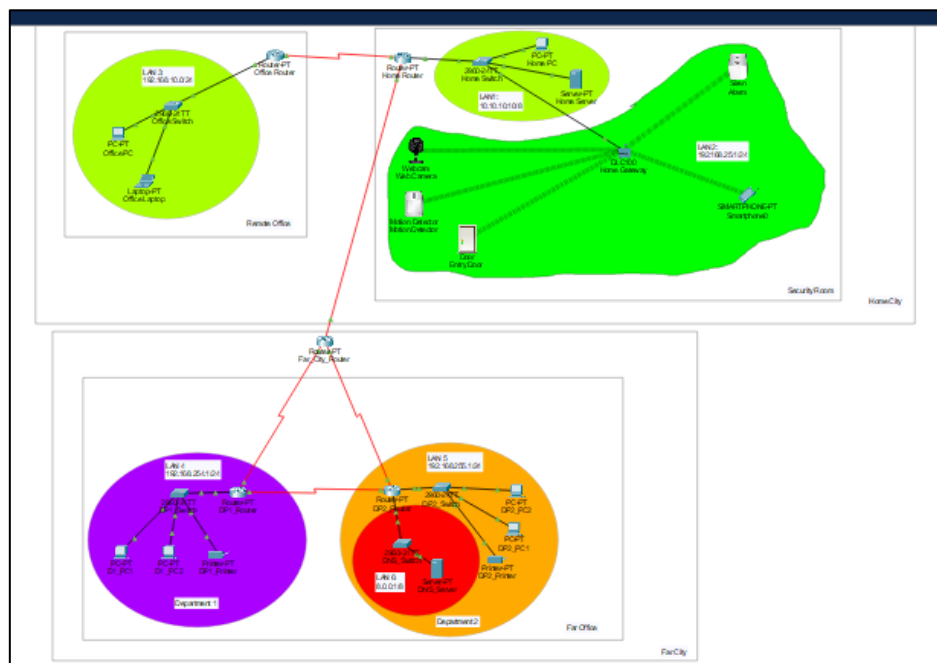


Figure 12, Remote Connection from Remote Far Office

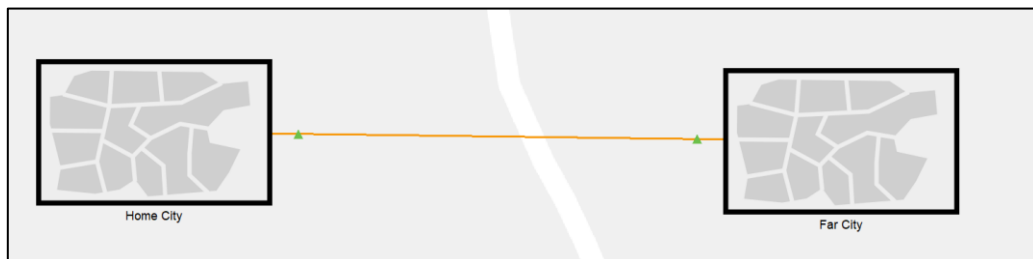


Figure 13, Connection between home city and far city Building in physical topology

iii. Implementing DNS Server:

The DNS Server in Department 2 has an IP address of 8.8.8.8 as shown in figure 14. We added the following addresses with fully qualified domain names as shown in figure 15.

www.dns.com → 8.8.8.8

www.home.com → 10.10.10.1 (home server containing IoT device information)

Now accessing this DNS Server, we must enter DNS server address in our hosts or end devices example is shown in figure 14.1.

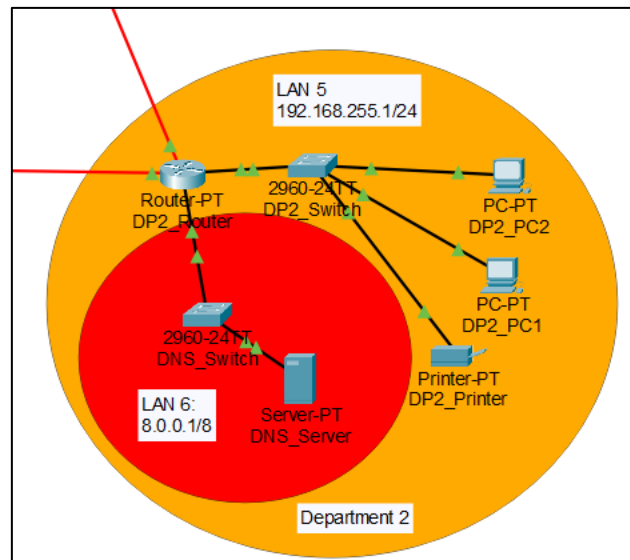


Figure 14, Implementation of DNS Server

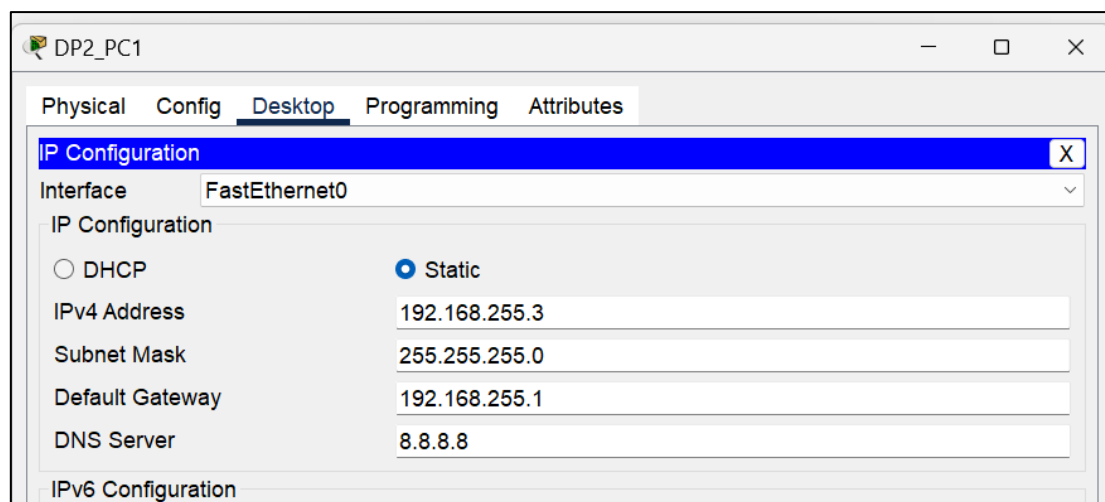


Figure 14.1, Entry of DNS Server in DP2_PC1

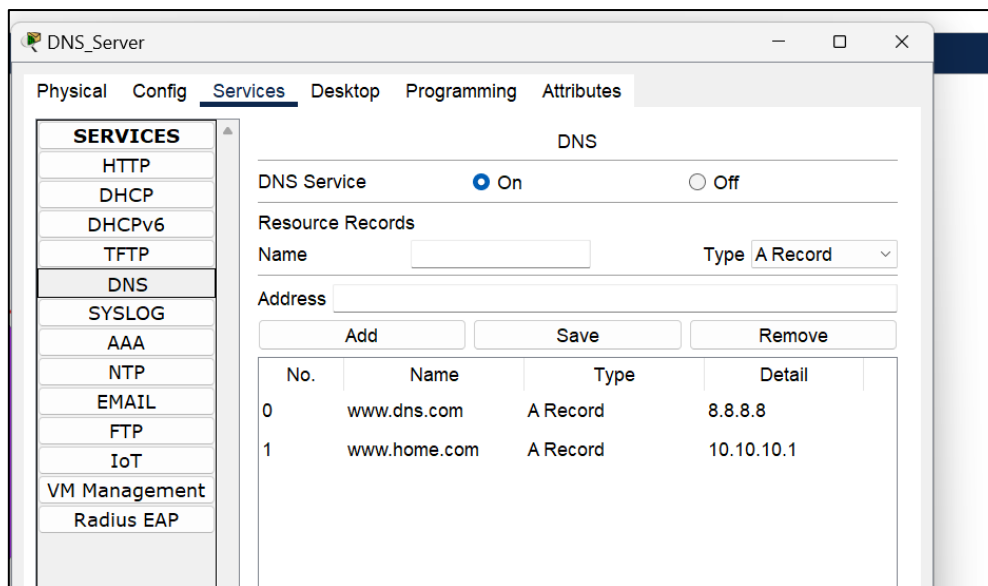


Figure 15, DNS Server Services

5. Results and Discussion:

The implementation of our smart security system using Packet Tracer yielded positive results, demonstrating the system's effectiveness in intrusion detection and prevention. We successfully set up and configured a network comprising IoT devices, a home gateway, and remote connections. The two scenarios—"No one detected" and "Intruder detected"—worked as intended. In the "No one detected" scenario, all security devices remained inactive, conserving resources and maintaining a low-profile operation. Conversely, in the "Intruder detected" scenario, the motion detector activated the webcam and alarm, effectively monitoring and responding to unauthorized entry attempts. Additionally, we extended the system's capabilities by enabling remote access from another building and city. The DNS server implementation allowed us to access the home server via both IP address and URL, providing flexibility and ease of access. This comprehensive setup verified the system's functionality in diverse environments and demonstrated the potential of IoT in enhancing security measures.

6. Conclusion:

This project successfully demonstrated the potential of IoT in developing advanced security systems using simulation tools like Packet Tracer. The smart security system we created is an effective solution for intrusion detection and prevention, capable of protecting homes, offices, rooms, or any restricted areas from unauthorized entry. By integrating various IoT components, implementing remote connections, and utilizing a DNS server, we extended the system's capabilities to ensure robust and user-friendly security solutions. This project underscores the importance of leveraging technology to develop comprehensive and adaptable security systems, addressing the growing demand for effective anti-theft solutions in a global context.

7. References:

Smart Security System using Packet Tracer | IoT application | Anti-Theft system using Packet Tracer. - YouTube